

RAVI KUMAR

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Summary

Network and systems engineer with 3+ years in public-sector and enterprise support, moving into cloud engineering. I operate a production Linux fleet I built myself — 229 scheduled jobs and 400+ systemd services on a Raspberry Pi 5 that serve real users and payments behind outbound-only Cloudflare Tunnels — with self-healing, failure paging, verified backups and fault-tested monitors. I build AWS infrastructure in Terraform with an apply → verify → destroy runner. Strengths: Linux, networking (DNS, DHCP, VPN, firewalls), troubleshooting, automation in Python, Bash and PowerShell, and writing runbooks people can follow.

Skills

- **AWS:** VPC, EC2, IAM, S3, ALB/Auto Scaling, ECS Fargate, CloudWatch + SNS, Budgets · **laC/automation:** Terraform, Bash, Python, PowerShell, Git/GitHub
- **Linux/ops:** Debian/Ubuntu, systemd units & timers, journald, cgroups/memory guards, nftables, backups & restores, incident response, runbooks, ITIL 4
- **Networking:** TCP/IP, DNS (Pi-hole, Unbound/DNSSEC, BIND, dnsmasq), DHCP, VLANs/802.1Q, Cisco switching, Palo Alto GlobalProtect, WireGuard, Tailscale, tcpdump/Wireshark, Cloudflare (Tunnel, Pages, DNS)
- **Endpoints/identity:** Active Directory, Microsoft Intune, MDM, macOS/Windows/Linux

Projects

- **Production fleet (Linux, systemd):** built and operate 229 scheduled jobs / 400+ services; every unit alerts on failure (OnFailure → Telegram), a 10-minute self-heal loop restarts with rate limits, a watchdog verifies the healers; nightly backups with verified restores; memory guards (earlyoom + cgroup caps) after a RAM-exhaustion freeze; detectors fault-injected before they ship. Public write-up: github.com/ravikus1457/fleet-ops.
- **Edge & DNS:** three public sites served from the fleet through outbound-only Cloudflare Tunnels (zero inbound ports, loopback-only origins); LAN DNS via Pi-hole → Unbound with DNSSEC; Tailscale mesh for operator access; WireGuard site-to-site to an AWS VPC.
- **PaperReel (paid product):** Stripe Checkout → signed webhook → systemd processing pipeline → signed download links; landing on Cloudflare Pages, API through a Tunnel; hard per-service spend caps in code; fault tests that can fail.
- **AWS labs (Terraform):** VPC, EC2 + ALB + Auto Scaling, IAM least privilege + S3, ECS Fargate, CloudWatch monitoring; run - lab . sh does init → validate → apply → exercise → destroy with evidence reports, a budget alarm and a tag-based orphan sweep. github.com/ravikus1457/aws-labs
- **Networking labs:** WireGuard site-to-site, subnetting/VLANs, DNS & DHCP (BIND/dnsmasq), nftables firewall & least-privilege ACLs, packet analysis.

Experience

Information Systems Technician I — Contra Costa County, Dept. of Public Works, Martinez, CA · Mar 2026 – Present - Diagnose and fix DNS, DHCP, Active Directory authentication and VPN routing failures across a distributed County network; document each resolution as a runbook. - Set up and configured Palo Alto GlobalProtect VPN access for County staff; troubleshoot GlobalProtect connectivity and routing. - Maintain IDF connectivity: patch and validate ports on Cisco routers and switches, confirm link status, document changes. - Manage County devices daily in Microsoft Intune (enrollment, profiles, compliance, remote actions); automate Active Directory audits and updates with PowerShell. - Led an end-to-end iPhone deployment for 125 employees: backups, Microsoft Bookings scheduling, rollout, training, issue resolution.

Information Systems Specialist I — Contra Costa County, IT Department, Martinez, CA · Nov 2025 – Mar 2026 - County-wide systems support: account administration, endpoint configuration, DNS/DHCP/AD connectivity resolution. Promoted to Technician I.

Junior Infrastructure Support Engineer (Contract) — Meta, Menlo Park, CA · Oct 2025 – Nov 2025 - Diagnosed routing, DNS, VPN, authentication and endpoint-performance issues in a high-availability production environment; partnered with engineering teams on escalations; supported Linux, macOS and Windows endpoints through MDM.

Cloud Operator — Bay Alarm, Concord, CA · Apr 2024 – Jun 2025 - Monitored cloud-connected enterprise devices for connectivity health and uptime; configured network devices and validated installations through cloud management consoles.

IT & Network Operations Analyst — Apple Inc., Elk Grove, CA · Feb 2023 – Jan 2024 - Troubleshoot TCP/IP, DNS, DHCP, VPN, authentication and MDM-profile issues across large-scale enterprise environments; documented repeatable resolution workflows.

Certifications & Education

- ITIL 4 Foundation (2026) · CompTIA Security+ · Network+ · A+ · Linux Essentials · ITF+
- In progress: AWS Certified Solutions Architect – Associate (SAA-C03), then AWS CloudOps Engineer – Associate
- B.S. Cloud and Network Engineering (AWS track) — Western Governors University, in progress (78 of ~120 units)
- IT Support Certificate — Year Up / Diablo Valley College, 2024